

DICOM Conformance Statement

Web DICOM Viewer

Zargantana / Web DICOM Viewer

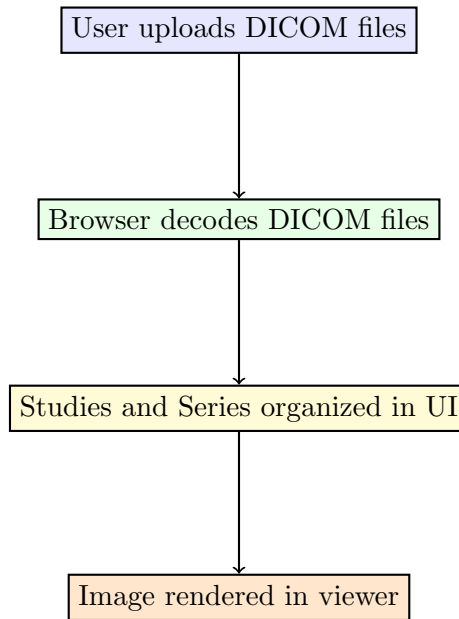
September 28, 2026

1. Introduction

This document is the DICOM Conformance Statement for the **Web DICOM Viewer**. This software component is designed for displaying DICOM images directly in a web browser. It supports uploading image files through a browser interface, visualizing them frame by frame with interactive navigation and rendering options.

2. Implementation Model

2.1 Application Data Flow



2.2 Functional Definition

The Web DICOM Viewer is a **VIEWER** application that does not establish network communication via DICOM protocol (no SCU/SCP roles). It operates solely on files provided through user upload.

Large files are read in ranges: only the Data Set up to the Pixel Data element is loaded to build the list of studies and series; each frame is read from the file when it is needed (locating it through the Extended or Basic Offset Table, or the fragment structure, for encapsulated Pixel Data) and compressed frames are decoded in background Web Workers. Decoded frames are kept in a memory cache of bounded size (256 MB on desktop computers, 128 MB on mobile devices). There is no file size limit other than those of the browser and the operating system,

except for Deflated Explicit VR Little Endian, which has to be inflated whole and is limited to about 512 MB per file.

3. AE Specifications

3.1 Web DICOM Viewer AE

- **SOP Classes Supported:**
 - All storage SOP classes containing image data (e.g., CT, MR, XA, US, NM, CR, OT).
- **Supported Transfer Syntaxes** (decoding only; all verified with a synthetic regression suite):
 - Implicit VR Little Endian (1.2.840.10008.1.2)
 - Explicit VR Little Endian (1.2.840.10008.1.2.1)
 - Encapsulated Uncompressed Explicit VR Little Endian (1.2.840.10008.1.2.1.98)
 - Deflated Explicit VR Little Endian (1.2.840.10008.1.2.1.99)
 - Explicit VR Big Endian, retired (1.2.840.10008.1.2.2)
 - JPEG Baseline, Process 1 (1.2.840.10008.1.2.4.50)
 - JPEG Extended, Process 2 & 4, 8 and 12 bit (1.2.840.10008.1.2.4.51)
 - JPEG Extended, Process 3 & 5, arithmetic coding, 8 and 12 bit, retired (1.2.840.10008.1.2.4.52)
 - JPEG Spectral Selection, Non-Hierarchical, Process 6 & 8 and 7 & 9, 8 and 12 bit, retired (1.2.840.10008.1.2.4.53, .54)
 - JPEG Full Progression, Non-Hierarchical, Process 10 & 12 and 11 & 13, 8 and 12 bit, retired (1.2.840.10008.1.2.4.55, .56)
 - JPEG Lossless, Non-Hierarchical, Process 14, any predictor, 2–16 bit (1.2.840.10008.1.2.4.57)
 - JPEG Lossless, Non-Hierarchical, First-Order Prediction, Process 14 Selection Value 1 (1.2.840.10008.1.2.4.70)
 - JPEG-LS Lossless and Near-Lossless (1.2.840.10008.1.2.4.80, .81)
 - JPEG 2000 Image Compression, Lossless Only and lossy (1.2.840.10008.1.2.4.90, .91)
 - JPEG 2000 Part 2 Multi-component, Lossless Only and lossy (1.2.840.10008.1.2.4.92, .93), except array-based multi-component transformations (see Limitations)
 - High-Throughput JPEG 2000, Lossless Only, with RPCL Options (Lossless Only) and lossy (1.2.840.10008.1.2.4.201, .202, .203)
 - RLE Lossless (1.2.840.10008.1.2.5)
- **Video Transfer Syntaxes** (decoding only, with the WebCodecs API of the browser; verified with synthetic files and a real H.264 study in Chrome and Edge). Some devices store a complete MP4 file (ISO/IEC 14496-12) in the Pixel Data instead of the elementary stream: the viewer reads its sample tables (non-fragmented MP4 only) and decodes it the same way; when the header has no Number of Frames or Frame Time, the values of the video are used.

- MPEG-4 AVC/H.264 High Profile / Level 4.1, BD-compatible, Level 4.2 for 2D and 3D video and Stereo High Profile, and their fragmentable variants (1.2.840.10008.1.2.4.102 to .106, and .102.1 to .106.1). The whole stream is read once, split into access units and decoded in order; each frame can be displayed, stepped forwards and backwards (decoding restarts at the previous IDR picture) and played as a cine loop.
 - HEVC/H.265 Main and Main 10 Profile / Level 5.1 (1.2.840.10008.1.2.4.107, .108), only where the browser and the computer support HEVC decoding (usually with hardware acceleration; verified on Windows in Chrome and Edge, where some hardware decoders do not accept pictures smaller than 320 by 240). Otherwise the viewer shows why the image cannot be displayed.
- **Retired and Private Transfer Syntaxes** (decoding only; verified with the same suite):
 - Papyrus 3 Implicit VR Little Endian, retired (1.2.840.10008.1.20)
 - GE Private Implicit VR Little Endian with Big Endian Pixel Data (1.2.840.113619.5.2)
 - Philips Private CT Explicit VR Little Endian, “CT-private-ELE” (1.3.46.670589.33.1.4.1)
 - PixelMed Private Encapsulated Raw Little Endian (1.3.6.1.4.1.5962.300.2)
 - Any other private or unknown Transfer Syntax whose Pixel Data contains a standard codestream: the codec is identified from the content (JPEG baseline, extended, lossless, progressive or arithmetic; JPEG-LS; JPEG 2000; HTJ2K; RLE; or uncompressed pixels of the expected size). Explicit or implicit VR is detected from the first data element.
 - **Supported Photometric Interpretations:** MONOCHROME1, MONOCHROME2, PALETTE COLOR (including segmented palette color lookup tables), RGB, YBR_FULL, YBR_FULL_422, YBR_PARTIAL_422, YBR_PARTIAL_420, YBR_ICT and YBR_RCT (JPEG 2000), and the retired HSV, ARGB and CMYK. Planar Configuration 0 and 1 are supported.
 - **Image Display Capabilities:**
 - Grayscale pipeline: Rescale Slope/Intercept (linear Modality LUT), VOI (Window Center/Width with the LINEAR, LINEAR.EXACT and SIGMOID functions, or a VOI LUT), and presentation (MONOCHROME1 and Presentation LUT Shape INVERSE are inverted).
 - When the file defines several windows, the user can select any of them; the first one is used by default. Without a window or VOI LUT, images of up to 8 bits are displayed over their full range and deeper images use the minimum and maximum of the frame.
 - The user can adjust the window interactively (dragging with the mouse, or with two fingers on touch screens: horizontal movement changes the width and vertical movement the center, as a LINEAR function applied to the output of the Modality LUT) and, for CT, choose predefined windows (brain, soft tissue, mediastinum, lung and bone). A user window replaces the window, VOI LUT or automatic window of the file until it is reset.
 - On touch screens, one finger steps through images or frames, two fingers zoom and pan, and a double tap fits the image again.
 - Pixel Padding Value / Range Limit: padding is shown in black and excluded from the automatic window.

- Supplemental Palette Color Lookup Tables (MONOCHROME2 with Pixel Presentation COLOR or MIXED).
- Multi-frame images are displayed frame by frame and played as a cine loop using Frame Time.

3.2 Supported Modalities

Modality Code	Description
CR	Computed Radiography
US	Ultrasound
XA	X-Ray Angiography
NM	Nuclear Medicine (e.g., gammagraphy)
OT	Other (Generic / Non-standard)
DX	Digital Radiography
CT	Computed Tomography (if uploaded)
MR	Magnetic Resonance (if uploaded)

4. Networking

Note: This application does not support DICOM network communication (no DICOM Upper Layer Protocol, DICOMweb, or DIMSE services).

5. Browser Interface and User Interaction

- Users upload DICOM files via an HTML file input element.
- The system displays a list of studies and associated series.
- Users can select a study and navigate through its images.
- Two display modes, chosen per series:
 - CT, MR and PT series with more than 10 images, and series of other modalities with more than 25 images: stack navigation with the mouse wheel.
 - Otherwise (always for US, CR, DX and XA): an image list with selection and, for multi-frame images, cine loop playback.

6. Limitations

- Not supported: JPEG 2000 Part 2 array-based multi-component transformations, hierarchical JPEG (processes 16–29), JPEG XL, MPEG-2 video (no browser decodes it), HEVC/H.265 video on browsers or computers without HEVC decoding, JPIP referenced pixel data, Float and Double Float Pixel Data, and ICC profile color management.
- JPEG Lossless with arithmetic coding (Process 15, 1.2.840.10008.1.2.4.58) is decoded on a best-effort basis for 8-bit images and has not been verified.
- Non-linear Modality LUT Sequences and per-frame functional group windows of enhanced multi-frame images are not applied.
- No support for DICOM C-FIND, C-MOVE, or other DICOM services.
- No integration with PACS or DICOMweb servers.

- Only file-based workflows supported.
- Proprietary compressions without a public decoder are recognized and reported to the user, with the advice to export the study in a standard Transfer Syntax, but not decoded: Sectra Compression (1.2.752.24.3.7.6), Sectra Compression LS (1.2.752.24.3.7.7), ALI Wavelet (1.2.840.113711.1.2.100.1), Algotec Compressed (1.2.840.113704.7.0.4.4), GE PACS COMPRESS_EXPRESS (2.16.840.1.113709.1.2.2) and PixelMed Bzip2 Explicit VR Little Endian (1.3.6.1.4.1.5962.300.1).

7. Security

The viewer operates entirely on the client side, without uploading data to a remote server. Security and confidentiality of patient data depend on user handling and browser environment.

8. IHE Integration Statement

This viewer does not currently support IHE Integration Profiles such as RAD-68 or XDS-I.b.

9. Support and Contact Information

Web DICOM Viewer

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